

Ride-Hailing Demand & Smart Charging for an Electric Fleet

Advanced Analytics and Applications

Team 04



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1 Problem description

Urban mobility is in the middle of a structural shift. Over the past decade, ride-hailing has grown from a niche offering into a force that is disruptively reshaping urban transport (Wang & Yang, 2019), changing travel behaviour and, in a number of cities, drawing riders away from public transit (Tirachini, 2020). At the same time, road transport is electrifying rapidly: extrapolating historical sales with a logistic growth model, electric vehicles are projected to reach roughly a third of the global passenger-car stock by the early 2030s (Rietmann et al., 2020). These two trends are converging, and established car manufacturers increasingly seek to move beyond selling vehicles towards operating mobility services themselves. This is where our **business goal** begins: a german car company is entering the space with a ride-hailing platform run on a fully electrified fleet, targeting the more receptive US market. Building vehicles and profitably operating a fleet of them are distinct competencies, and running an on-demand service requires a quantitative understanding of where and when demand arises, which the manufacturer does not yet possess.

That understanding matters because demand for on-demand mobility has pronounced spatial and temporal structure (Ke et al., 2017) and is further shaped by exogenous conditions such as weather. An operator must therefore decide on three levels. Strategically, which districts of a city to enter first; tactically, how many vehicles to deploy and at which times and places; and operationally, how to position and, since the fleet is electric, charge those vehicles efficiently. Answering these questions requires demand data at fine spatial and temporal resolution, which does not yet exist for a platform that has not launched. As a proxy, we use the public taxi trip records of the City of Chicago from 2024 onwards: taxis are a reasonable stand-in for on-demand ride-hailing, and Chicago serves as a representative large US city. Nevertheless, the proxy is imperfect, as taxis differ from app-based hailing in booking behaviour and market coverage, a limitation we keep in view when drawing conclusions.

From this business problem we derive our **data-mining goal**, a chain of five analytics tasks that build on one another. First, *data preparation* turns the large, privacy-restricted raw trip records into demand panels: we validate and clean the records, discarding implausible or inconsistent trips, discretise the city with a hexagonal H3 grid, and enrich each cell and time window with external weather data. Second, *descriptive spatial analytics* characterises how demand varies across space and time, selects an appropriate spatial and temporal resolution, and identifies persistent demand hotspots using Gaussian Mixture Models and kernel density estimation. Third, *predictive analytics* estimates the expected demand per spatial unit and time window as a function of calendar, weather and location, using Support Vector Machines and feed-forward neural networks; the aim is conditional estimation, quantifying how demand depends on observable conditions, rather than autoregressive forecasting of the next interval from the current one. Fourth, *prescriptive analytics* addresses the electric dimension by designing a home-charging agent through reinforcement learning that minimises recharging cost while ensuring enough energy for the working day. Finally, *synthesis* consolidates these results into recommendations for the operator, including whether it should rely on private or public charging infrastructure. Together, the five tasks move from describing demand, to explaining it, to acting on it.

2 Data description

We draw on three datasets. The primary source is the *Taxi Trips (2024-)* record published by the City of Chicago on its open data portal, holding 14,802,849 trips across 23 fields. It spans 1 January 2024 to 31 March 2026 and covers 3,735 vehicles, 46 licensed taxi companies and all 77 of Chicago’s community areas. To this we add historical weather for Chicago from

the Open-Meteo API, at hourly resolution (19,728 observations of temperature, precipitation, rain, snowfall, wind speed and a categorical WMO weather code) alongside 822 daily aggregates that add sunshine duration and the number of precipitation hours. A GeoJSON of the 77 community-area boundaries supplies the geometry used to map and aggregate trips spatially. We decided against point-of-interest extensions, as the records already at hand cover the questions set out in Section 1, so the expected analytical gain did not justify the additional integration effort.

Each row represents one completed taxi trip. Location is reported at three nested levels, census tract within community area, plus a centroid coordinate pair. The 23 fields group as follows, with the full field-by-field dictionary given in the Appendix; for demand modelling only the pickup-side time and location fields are required.

Table 1: Structure of the raw Chicago taxi trip records: the 23 fields grouped by type, with the defining characteristics of each group.

Group	Fields	Notes
Identifiers	Trip ID, Taxi ID, Company	3,735 vehicles, 46 companies
Time	Trip Start / Trip End Timestamp	rounded to 15-minute bins
Trip metrics	Trip Seconds, Trip Miles	duration and distance
Space	Pickup / Dropoff Census Tract, Community Area, Centroid Latitude, Longitude and Location	coordinates are area centroids, not exact points
Money	Fare, Tips, Tolls, Extras, Trip Total	0.21% missing
Payment	Payment Type	

Two structural properties shape everything that follows. First, all timestamps are rounded to quarter-hour bins, taking only the minute values 0, 15, 30 and 45, which sets a hard floor on the temporal resolution any analysis can achieve. Second, location is restricted for privacy rather than by measurement: no exact coordinates are published, and the coordinate fields give the centroid of the enclosing area. Tract identifiers are further suppressed for the majority of trips, missing in 55.65% of pickups and 56.98% of dropoffs, whereas the community area is far more complete, missing in 2.79% and 8.92% of cases. Because suppression applies wherever a combination of tract and time holds too few trips to guarantee anonymity, it removes precisely the sparse observations and is therefore correlated with demand density. Analysis at tract level would be biased towards dense areas, which is why the choice of spatial unit in the next section cannot simply default to the finest available geography.

Two limitations remain. The raw records contain recording artefacts, most visibly trips of zero distance or zero duration, which we quantify and address in the next section. More fundamentally, the dataset captures served demand only: requests that went unmet leave no trace, so the data understate true demand in exactly those places and periods where supply was scarce.

3 Data preparation

The raw records are turned into an analysable demand panel in three stages: cleaning, discretisation and aggregation. Syntactic cleaning harmonises data types and parses timestamps,

and no fully duplicated rows occur. Semantic validation then distinguishes two kinds of implausibility, and that distinction carries the whole preparation. Records that cannot describe a real journey are excluded outright: 98 trips end before they start, 12,104 last longer than three hours, and 1,202 exceed 100 miles. Records implausible in only one attribute are retained as demand events but withheld from the statistics they would distort, namely 1,294,820 trips of zero distance, 201,927 of zero duration, and fares above 200 dollars. A zero-distance record still marks a real pickup: deleting it would understate demand, while averaging it would corrupt the distance statistics, so it counts towards demand and is suppressed only where it would mislead. The complete set of validation rules and thresholds is documented in the Appendix.

Missing values are likewise not treated globally. Since the demand model describes where trips begin, missing dropoff information never removes an observation. Each derived dataset then carries its own requirement, so every analysis runs on the largest subset valid for it: the hexagon panel needs pickup coordinates, while the community-area panel, which we retain for the human-readable views in Section 4, needs the community area. Requiring pickup coordinates removes 405,819 trips, which together with the hard exclusions leaves 14,382,862 of 14,802,849 records, or 97.2%. We checked whether this distorts the temporal picture. Across months the missing rate is stable between 2.4% and 3.0%, but across the day it is not: it falls to 1.9% in the late afternoon and rises to 6.8% around 4 a.m. This is the density-driven mechanism from Section 2, as fewer night-time trips push more area and time combinations below the anonymity threshold. The effect is small in absolute terms, but night-time demand is marginally understated, which we revisit when interpreting results.

For the spatial unit we compared Chicago’s community areas with a hexagonal grid. Community areas are administratively meaningful and human-readable, which makes them valuable for interpretation, but they vary considerably in size and shape, so demand densities are not comparable across them. We therefore discretise the city with the H3 indexing system, developed by Uber for exactly this purpose, which provides cells of uniform size and shape at a tunable resolution. We construct resolutions r4 to r8, where a higher index means smaller cells and each step divides the area by roughly seven. Two independent criteria bound the sensible range from both sides, as Table 2 shows. Downwards, r4 and r5 leave only 4 and 8 cells for the entire city, three of which already carry 80% of all demand, far too coarse to guide where vehicles should be deployed. Upwards, two things break at once. Sparsity rises steeply, with the share of empty cell and window combinations climbing from 12.6% at r6 to 49.2% at r7 and 82.0% at r8, so the panel increasingly records absence rather than demand. More fundamentally, measurement validity fails: because coordinates are area centroids of uneven precision, with trips of known tract carrying one of 607 tract centroids and suppressed trips one of merely 77 community-area centroids, a hexagon smaller than its reporting area cannot observe real structure. At r8, 67% of occupied hexagons are fed by a single centroid, so their apparent detail only reproduces where centroids happen to lie. Resolution r6 is the finest level at which both criteria remain intact, at more than three times the spatial detail of r5, and we adopt it as the canonical unit.

Table 2: Spatial resolution sweep at fixed four-hour windows. Coarse resolutions leave too few cells to guide deployment, while from r7 onwards sparsity and measurement validity deteriorate together, identifying r6 as the canonical unit.

Resolution	Cell area	Cells	Empty share	Single-centroid cells
r4	1,770.3 km ²	4	1.8%	0%
r5	252.9 km ²	8	1.0%	0%
r6	36.1 km ²	27	12.6%	0%
r7	5.2 km ²	128	49.2%	20%

Resolution	Cell area	Cells	Empty share	Single-centroid cells
r8	0.7 km ²	475	82.0%	67%

The temporal grid is bounded from below by the quarter-hour rounding of the source timestamps. We build one-hour, four-hour, eight-hour and daily windows and adopt four hours as canonical, since hourly bins reproduce the same sparsity problem while daily bins average away the within-day pattern on which any deployment decision depends. The weather records are aggregated to these windows and joined on the time key, contributing temperature, rain, snowfall, total precipitation, wind speed and the categorical weather code, so that demand variation driven by conditions is not misattributed to the calendar.

The aggregation returns one row per hexagon and time window, carrying the number of pickups as the target variable alongside mean trip distance, duration and fare, and the number of distinct taxis active in the cell. The window timestamp is retained as the basis for the cyclical calendar features, namely time of day, day of week and month, that enter the models in Section 4. Separately, we derive idle time per vehicle as the gap between one trip ending and the same taxi starting its next, which supports the utilisation view in Section 4. One step deserves emphasis. A grouped aggregation produces rows only where at least one trip occurred, so a cell without pickups is absent rather than zero, and a model trained on that panel would never observe the quiet half of the distribution and would systematically overestimate demand. We therefore complete the panel across the full Cartesian product of cells and windows and set the missing combinations to zero, leaving trip characteristics undefined because no trips exist to average. The canonical dataset comprises 27 hexagons across 4,927 four-hour windows, that is 133,029 observations, of which 12.6% are genuine zeros introduced by this step alone. Figure 1 makes the trade-off visible.

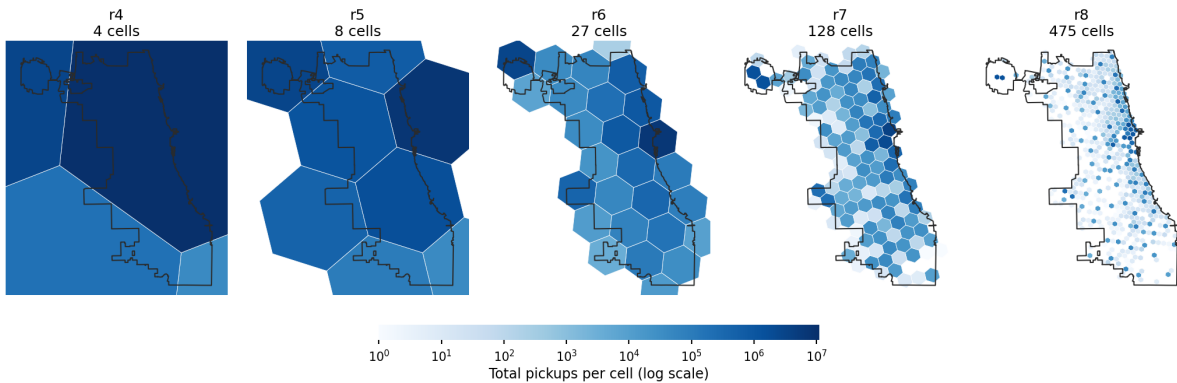


Figure 1: Total pickups per cell across H3 resolutions r4 to r8, on a shared logarithmic colour scale, with the city boundary for orientation. Coarse grids collapse the city into a handful of cells, while at r7 and r8 the surface fragments into largely empty ones. Resolution r6 is the finest level that still resolves spatial variation without dissolving into noise.

4 Data analytics

4.1 Descriptive analytics

City-wide demand follows a stable and highly predictable rhythm. Weekdays are markedly busier than weekends, averaging 18,922 against 13,919 trips per day, and both peak in the late

afternoon at 5 p.m. Their shapes differ in a way characteristic of commuting, as Figure 2 shows: on weekdays the hours from 6 to 9 a.m. carry 18.5% of the day’s trips, against only 10.5% at weekends, while the weekend redistributes demand into the small hours, where midnight to 4 a.m. accounts for 6.8% of trips against 3.1% on weekdays. Weather adds a genuine secondary effect on top of this calendar structure. Demand rises steadily with temperature, from roughly 2,400 trips per four-hour window in the coldest decile of observations to 4,142 in the warmest, and rain lifts demand by a consistent 5 to 7% (see Figure 8 and Figure 10). Neither association is an artefact of mild weather coinciding with busy hours, as both persist when windows are compared within the same time of day. Snowfall, by contrast, shows no reliable effect in our sample, since windows with heavy snow are too rare to support a conclusion (see Figure 9). We read these patterns as commuting producing the weekday morning shoulder, business and nightlife density sustaining the long afternoon plateau, leisure travel shifting the weekend into the night, and the weather response reflecting substitution away from walking and public transit in unpleasant conditions.

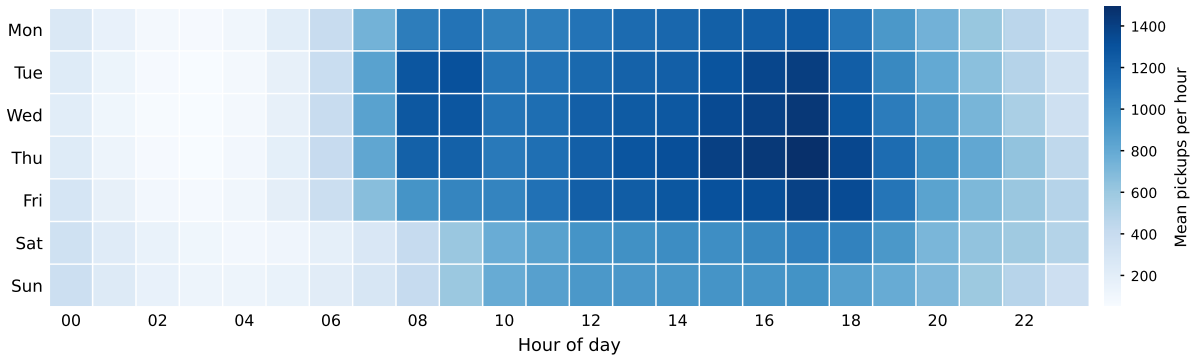


Figure 2: Mean pickups by weekday and hour of day. The weekday morning shoulder from 6 to 9 a.m. is absent at weekends, while weekend demand extends further into the night. Both patterns peak in the late afternoon.

Spatially, demand is extremely concentrated: at the canonical resolution a single downtown cell accounts for 46.8% of all pickups, and together with O’Hare for 67.5%. The more consequential finding is qualitative, as the two cores operate in different ways. The downtown core generates a high throughput of short trips, whereas the airports generate fewer but considerably longer and more valuable ones, with O’Hare averaging 14.2 miles per trip and the highest tip rate in the city at 19%. Origin-destination flows are asymmetric, so vehicles accumulate at destinations and must be repositioned empty, and idle time is highest in peripheral areas and low-demand hours, marking local over-supply.

To test whether these cores are genuine rather than an artefact of our own grid, we identified hotspots without reference to any grid, fitting a Gaussian mixture model to the raw pickup coordinates with the number of components selected by the Bayesian information criterion, and adding a kernel density estimate for the continuous density gradient (see Figure 6 and Figure 7). The component centres fall on the same cores, the Loop, Near North, O’Hare and Midway, confirming them as real features of the demand surface rather than products of our discretisation. Comparing these grid-free centres with the leading cells of each hexagon resolution also shows how the grid distorts them: r4 merges the Loop and Near North into one cell, while r8 splits a single airport across several. Resolution r6 aligns most closely with the mixture centres, which corroborates the choice made in Section 3 on entirely independent grounds.

4.2 Predictive analytics

We now estimate demand for each spatial unit and time window. The models describe demand as a function of observable conditions rather than extrapolating it from recent demand, because the operational question is what demand to expect under given circumstances, not what follows mechanically from the preceding interval. The target is the pickup count per hexagon and four-hour window. It is heavily right-skewed and contains 12.6% zeros, so we model it on a log scale and invert the transformation before scoring. The predictors are the cyclical encodings of time of day, day of week and month, the cell centroid coordinates, the great-circle distances to the three demand centres identified in Section 4.1, and the four-hourly weather variables. Representing location as continuous coordinates and distances rather than one indicator per cell allows both models to share information between neighbouring units and to generalise across space, which is what a location-conditioned question requires. Full feature definitions are given in the Appendix.

The validation strategy is built around one specific leakage risk. The four-hour windows of a single day share weather, calendar values and unobserved local shocks, so a random split would place some windows of a day in training and the rest in test, letting a model exploit day-specific signal that its features do not contain. We therefore split by whole calendar days, so that a day appears entirely in training or entirely in test. A grouped random sample of 15% of days forms an untouched holdout, tuning uses grouped cross-validation on the remaining development set, and standardisation, the target transform and the hotspot anchors are fitted on training data only and refitted within each fold. We report three measures. MAE gives the average error in trips and is the most interpretable, RMSE penalises the large misses that occur at demand peaks, and R^2 is scale-free. Only R^2 remains comparable when the resolution changes, for reasons set out below, so we lead the comparison with it.

We begin with a linear support vector regressor as a kernel-free baseline, add an RBF kernel selected by grid search, and compare both against a feedforward neural network in a baseline and a tuned configuration. Table 3 reports holdout performance. The linear model explains 63% of the variance in demand, which is inadequate and expected, since it cannot represent the interaction between location and time of day that drives the pattern. The RBF kernel raises this to 88%, and the neural network to 93%, with tuning reaching 96% and roughly halving the average error in trips relative to the kernel machine.

Table 3: Holdout performance at H3 r6 and four-hour windows. The benchmark predicts the historical mean demand for each cell and time of day, fitted on the development set only.

Model	R^2	MAE	RMSE	Inference
Linear SVM	0.633	65.56	203.49	0.03 s
RBF SVM	0.875	30.83	118.86	29.18 s
Neural network, baseline	0.926	26.75	91.36	0.01 s
Neural network, tuned	0.957	19.35	69.93	0.01 s
Naive benchmark (historical mean)	0.898	29.14	107.13	

Whether this complexity is justified is decided by comparison with a trivial alternative. A lookup table that simply returns the historical mean demand for each cell and time of day, with no learning at all, reaches R^2 0.898. The RBF support vector machine, after a full grid search, reaches 0.875 and therefore fails to beat this lookup on either measure. Only the neural

network adds genuine signal, reaching 0.957 and reducing the average error by a third relative to the benchmark. The computational comparison points the same way: the kernel machine needs 29 seconds to score the holdout, because kernel methods scale roughly quadratically in the number of training samples, whereas the network needs 0.01 seconds, a factor of about two thousand, and it trains on the full development set rather than a subsample. Interpretability is not the trade-off it is commonly assumed to be, since a SHAP analysis (see Figure 14) shows the distances to the demand centres and the time-of-day terms dominating the prediction, which is a defensible explanation for a non-technical audience. The deep-learning approach is therefore warranted here, not because it is more elaborate, but because it is the only model that improves on a lookup table while also being cheaper to serve.

Performance across resolutions must be read with care, because raw MAE falls mechanically as cells become finer and hold fewer trips. Census tract achieves an MAE of 1.5 against 62.5 at r5, yet is by far the weaker model, which is why the comparison rests on R^2 . Refitting both models at each resolution exposes the granularity and sparsity trade-off directly. Spatially, performance degrades as the grid tightens: the network reaches R^2 0.945 at r5 and 0.942 at r6, falls to 0.775 at r7, where 49% of observations are empty, and collapses to 0.375 at census tract level. Community areas sit between r6 and r7 on every dimension, with 77 units against 27 and 128, a 17.3% zero rate against 12.6% and 49.2%, and R^2 0.922 against 0.942 and 0.775, which makes them a defensible alternative wherever a human-readable unit is preferred. Census tract fails for two compounding reasons: the grid is extremely fine, with mean demand of 2.1 trips per window, and, as established in Section 2, only 45.6% of trips carry a tract identifier while the suppressed ones are precisely the sparse observations. The unit is therefore unsuitable, not because the models are weak but because the input is simultaneously too fine and structurally incomplete in a demand-correlated way. Temporally the direction reverses, with R^2 rising from 0.927 at hourly to 0.958 at daily windows, simply because aggregation averages out noise. This is not an argument for daily buckets, since a daily figure cannot distinguish the vehicles needed at 8 a.m. from those needed at 6 p.m. Full tables for both sweeps are given in the Appendix.

Three limitations qualify these results. The first concerns the weather variables and is a direct consequence of the temporal resolution. Aggregating weather to four-hour windows attenuates intermittent conditions far more than smooth ones: temperature varies by only 3.2 °F within an average window and survives aggregation almost intact, whereas in windows recording rain only 58% of the constituent hours were actually wet, and in 35% of them the rain fell within a single hour. A window labelled as wet is therefore frequently dry for most of its duration. This measurement error biases the estimated contribution of precipitation towards zero and explains why the weather terms carry little weight in the model even though Section 4.1 establishes a real association. Finer bins would measure weather more faithfully but would make the demand panel too sparse to learn from, as the zero rate doubles from 12.6% to 26.8% at hourly resolution, so the trade-off cannot be resolved within a single resolution. Second, the target is served demand, so the models reproduce the historical supply-constrained pattern and understate demand wherever vehicles were unavailable. Third, the panel contains trip characteristics such as mean distance and fare that are known only after the trips have occurred; these are excluded to avoid leakage, at the cost of some predictive signal. The most promising improvements follow from these gaps: point-of-interest and event data to explain the residual variation around venues and airports, an explicit treatment of unmet demand, and a weather representation that preserves intensity within the window, for example the wet fraction of the window rather than its mean.

4.3 Smart Charging Using Reinforcement Learning

4.3.1 MDP and model setup

The assignment’s two-hour home-charging window is divided into eight 15-minute decisions. The state contains the slot and battery charge; actions are 0, 7, 14, or 22 kW, and 60 kWh is chosen as the battery-capacity parameter. Energy demand is modelled as $\max(0, D)$, where D is normal with mean 21.4 kWh and standard deviation 14.1 kWh. These parameters are derived from pooled taxi-day distances using an assumed conversion of 0.40 kWh/mile. Slot cost is $\alpha_t * (\exp(0.1 * \text{power}) - 1)$. A shortfall incurs 20 plus 9.54 per missing kWh. The first day starts at 5 kWh; as an extension to the single-day setup, later days start with the preceding day’s remainder.

4.3.2 Learning approach

Tabular Q-learning uses the slot and battery charge, snapped to a 0.5 kWh grid, with the four charging actions. It is trained for 200,000 simulated days using epsilon-greedy exploration; epsilon decreases from 1.0 to 0.02 and the learning rate decays. Future rewards are undiscounted within a day and discounted by 0.95 at the day boundary. Dynamic programming (DP) uses the same action set and battery grid and converges after 187 iterations. However, DP also approximates demand on an 81-point grid that does not reproduce the simulator’s clipped normal distribution exactly. DP is therefore an approximate model-based reference, not proof of the exact optimum in the simulation environment.

4.3.3 Results and comparison

All policies are evaluated for 5,000 days using the same random seed and hence the same demand sequence. The DP policy has the lowest average charging-cost value, 1.0521, and the second-lowest shortfall rate, 0.52%. Q-learning reaches 1.3191 and 3.02%. Constant-medium charging reaches 1.2844 and 3.90%, so, based on the reported values, Q-learning reduces shortfall by 0.88 percentage points for 0.0347 additional charging cost. Maximum charging has the lowest shortfall rate, 0.46%, but costs 3.3737; the cheapest-slots heuristic reaches 2.1106 and 1.44%. Thus, Q-learning improves reliability over constant-medium charging but does not dominate all baselines. Because the output omits penalty-inclusive return, it supports a cost-reliability comparison rather than a complete ranking by the training objective.

4.3.4 Implications and limitations

The sensitivity analysis re-solves DP for each setting and evaluates each resulting policy over 2,000 days. Increasing demand’s standard deviation from 5 to 20 kWh raises average charging cost from 0.8056 to 1.3114 and shortfall from 0.10% to 3.60%. Raising both penalty components from 0.5 to 4 times their base values increases charging cost from 1.0460 to 1.1248 and reduces shortfall from 0.45% to 0.25%. Tested battery capacities of 40, 60, and 80 kWh produce observed shortfall rates of 9.20%, 0.15%, and 0.00%, respectively. These are model outcomes, not dollar costs or real-world forecasts. They depend on the chosen penalty, assumed energy conversion, clipped normal demand, stylised prices, fixed charging window, and omission of charging losses, public charging, battery degradation, and fleet interactions.

5 Discussion & Outlook

5.1 Implications for the fleet operator

For the client, the main implication is that market entry should start with a focused operating area rather than immediate city-wide coverage. The cleaned Chicago taxi data, enriched with weather data and aggregated to H3 r6 x 4h, shows that demand is strongly structured by time and location. Central areas and airport-related zones require different operating logic: dense downtown areas support frequent short trips and high turnover, while airport trips are longer, more range-sensitive, and operationally more demanding. Since taxis are only a proxy for ride-hailing demand, these findings should not be transferred blindly to another market, but the pipeline can be reused for the actual target city.

Strategically, the operator should first cover persistent demand centres and the corridors between them. Tactically, the tuned neural network is the preferred demand model because it performs best against the historical-mean benchmark, while H3 r6 x 4h offers a practical compromise between operational detail and sparsity. However, the prediction target is trip demand, not fleet size. Fleet requirements still depend on trip duration, idle time, vehicle productivity, target service levels, repositioning, and charging downtime.

5.2 Further analyses using the existing data

The most important next step is to translate predicted demand into vehicle requirements. The available taxi data already contains trip duration, idle time, and taxi identifiers, which could be used to estimate how many trips one active vehicle can realistically serve per area and time bucket. Combining this productivity estimate with predicted demand would provide a first fleet-sizing model, including reserve capacity for peaks and prediction errors.

A second extension is a simple fleet simulation. Origin-destination flows can be used to model where vehicles accumulate, where shortages occur, and how much empty repositioning is needed. This would connect the descriptive flow analysis with the predictive demand model and the smart-charging component. Instead of only asking where demand occurs, the operator could evaluate whether enough vehicles are available in the right places at the right times.

Further model validation should focus on robustness. Rolling time-based validation, prediction intervals, and residual analysis would show whether the model remains reliable across seasons, weather conditions, and unusual demand periods. This matters because operational planning depends not only on average performance, but also on how badly the model fails in peak situations.

5.3 Additional external data sources

Several external data sources could improve the analysis. Point-of-Interest data from OpenStreetMap would be especially useful, because offices, airports, transit stations, hotels, residential areas, and entertainment venues help explain why demand differs between spatial cells. Event calendars, public holidays, traffic speeds, road closures, and public-transport disruptions could further improve demand and travel-time predictions.

For the charging part, the current model should be extended with real vehicle and infrastructure data. Relevant sources include vehicle telematics, battery state-of-charge records, energy consumption by trip type, home or depot availability, electricity tariffs, grid constraints, and public charger data. Public charger data should include location, power level, price, utilisation,

waiting times, and outages. Without this information, the charging model should be interpreted as a controlled proof of concept rather than a final infrastructure decision.

5.4 Charging infrastructure recommendation

The results support a hybrid charging strategy. Private charging should form the operational backbone because it offers more control over timing, availability, and cost. The smart-charging model shows that scheduled charging can maintain high reliability without always using maximum power. This is especially valuable for vehicles with predictable home or depot charging windows.

Public charging should complement private charging rather than replace it. It is useful as a reserve option for unexpectedly high energy demand, low starting charge, vehicles operating far away from private chargers, or peak-demand days where normal charging windows are insufficient. However, public charging introduces uncertainty through detours, waiting times, charger availability, and price variation, none of which are fully captured in the current model.

Therefore, the client should not decide immediately between fully private or fully public charging. A sensible next step is a pilot in the target market. The pilot should collect local ride-hailing demand, vehicle utilisation, energy consumption, charging behaviour, and public charging reliability data. Based on this evidence, the operator can size a private charging backbone and define how much public charging capacity is needed as flexible backup.

6 Citations and References

You can easily cite your sources using BibLaTeX. For example, you can cite the Quarto documentation (Team, 2024) or a textbook like “R for Data Science” (Wickham et al., 2023). The bibliography will be automatically generated at the end of the document.

7 Conclusion

The conclusion should summarize your findings and suggest future work.

8 Appendix

8.1 H3 hexagon area corresponding to specific resolution

Resolution	Avg Area	Rough Scale
R4	~1,770 km ²	Large city / metro region
R5	~253 km ²	City district
R6	~36 km ²	Large neighborhood
R7	~5.2 km ²	Neighborhood
R8	~0.74 km ²	A few city blocks

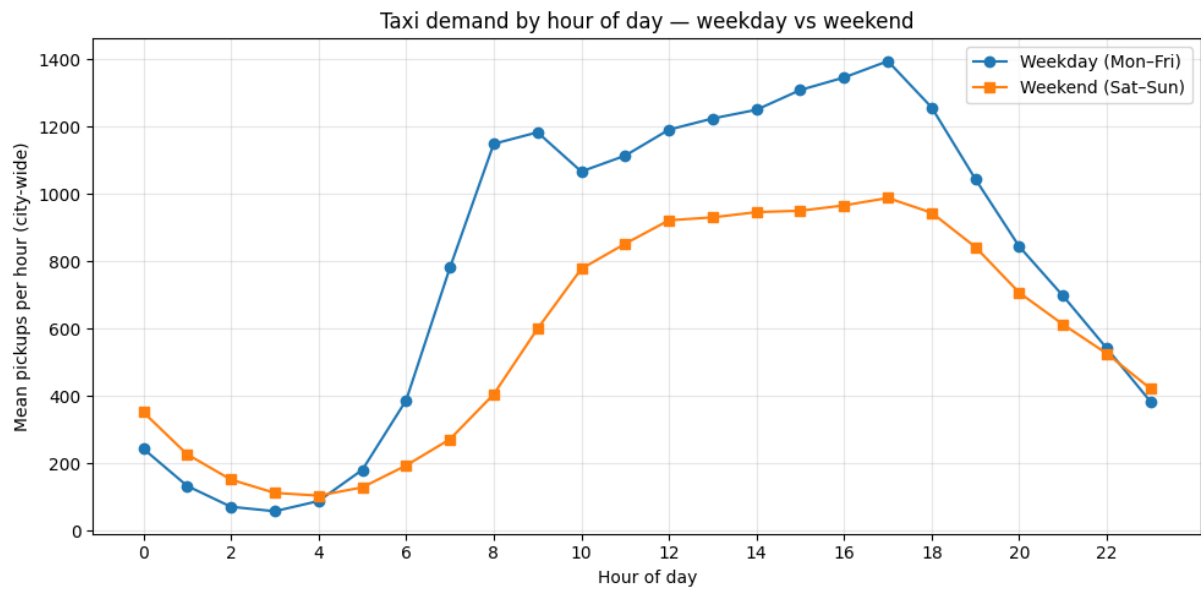


Figure 3: Taxi demand by hour of day - weekday vs weekend

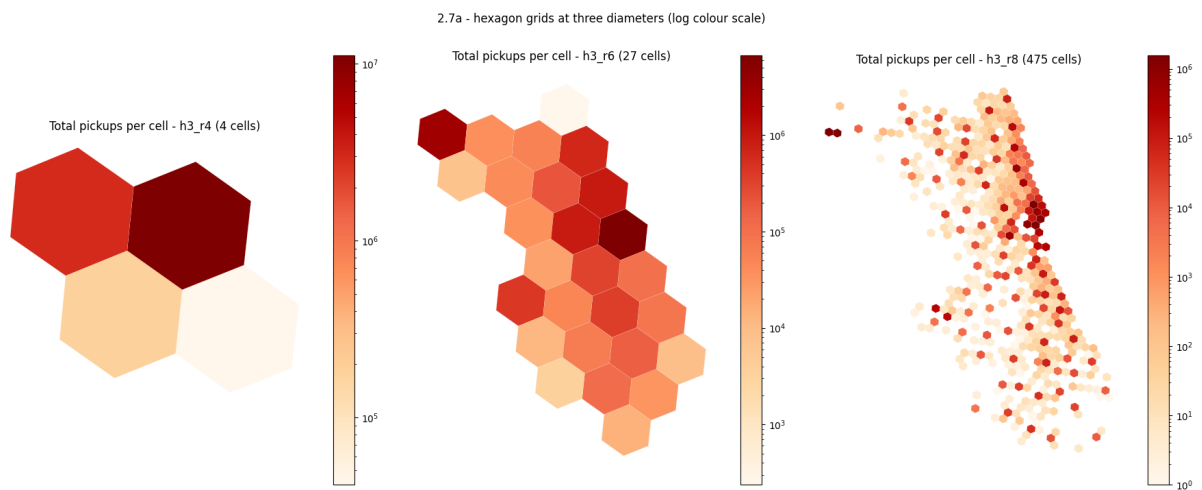


Figure 4: Total pickups per cell

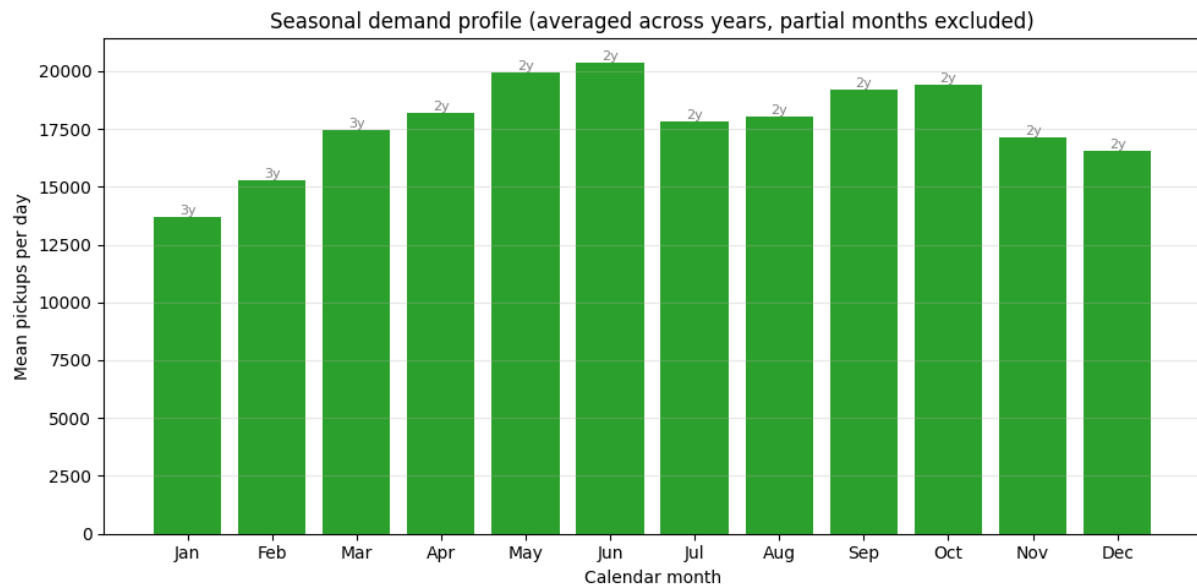


Figure 5: Seasonal demand profile

8.2 Baseline Neural Network Architecture

8.3 candidate values of hyperparameters for tuning Neural Network model

Parameter	Candidate Values
hidden_sizes	(64, 32), (128, 64, 32)
dropout	0.1, 0.2
lr	1e-3, 5e-4
batch_size	256, 512

8.4 Baseline Neural Network model result

8.5 Best Neural Network model

8.6 Feature importance - SHAP - of best Neural Network model

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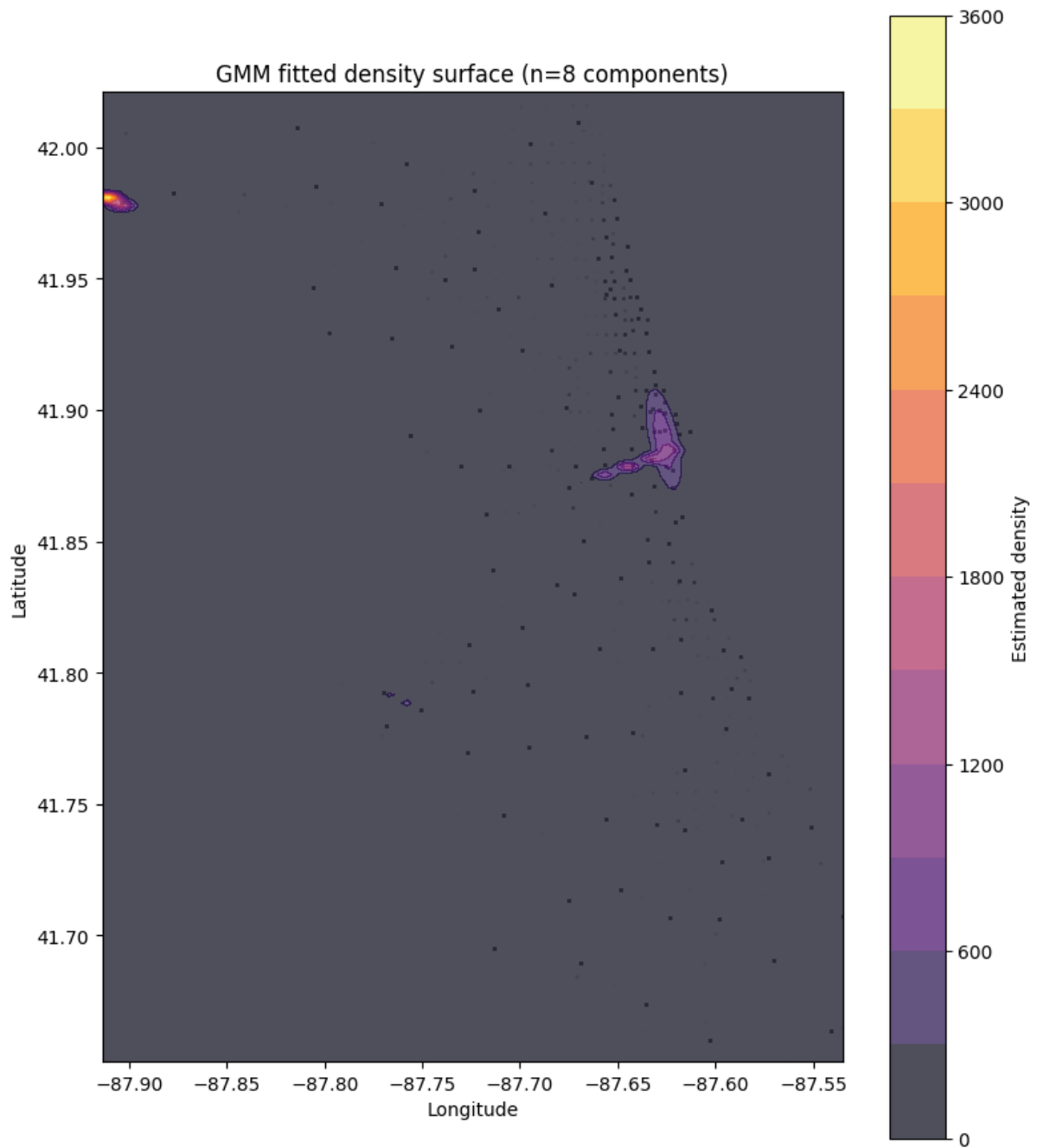


Figure 6: GMM

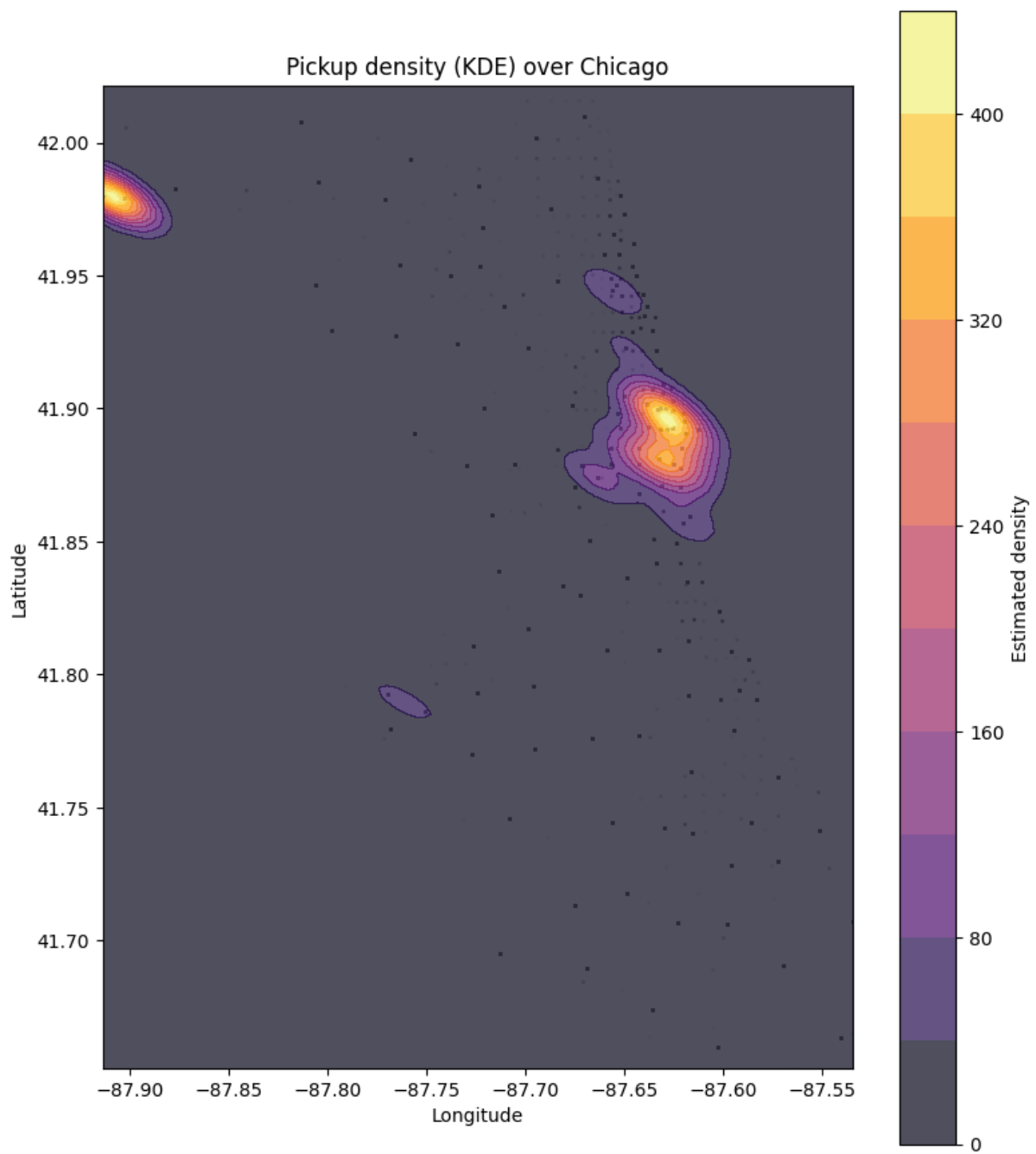


Figure 7: KDE

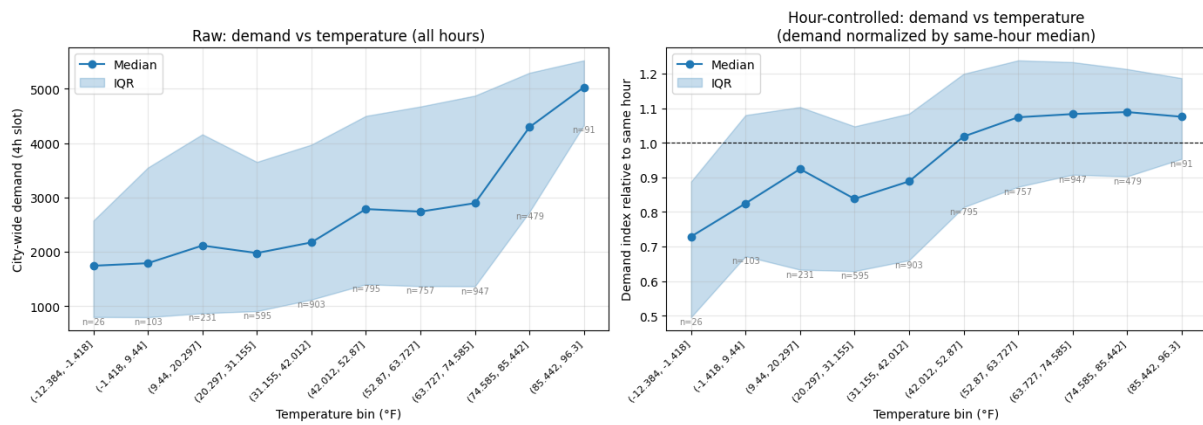


Figure 8: Demand vs temperature

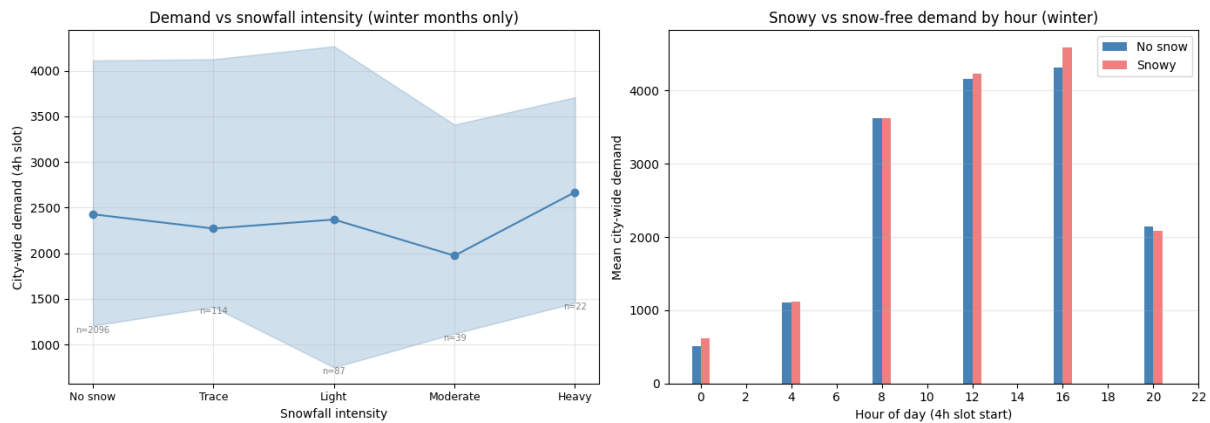


Figure 9: Demand vs snow intensity

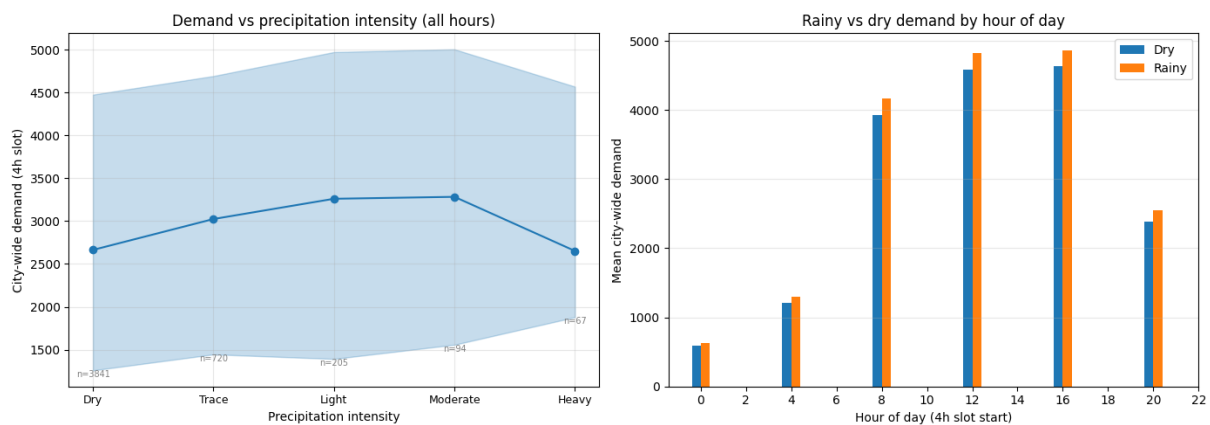


Figure 10: Demand vs precipitation intensity

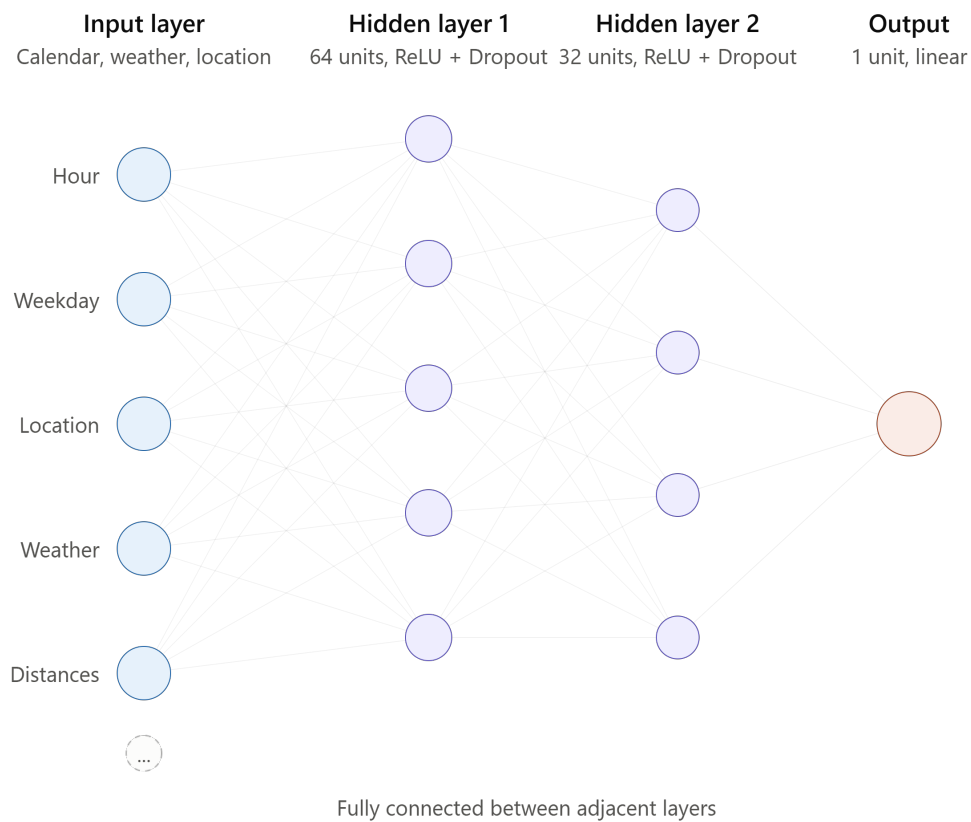


Figure 11: Model architecture

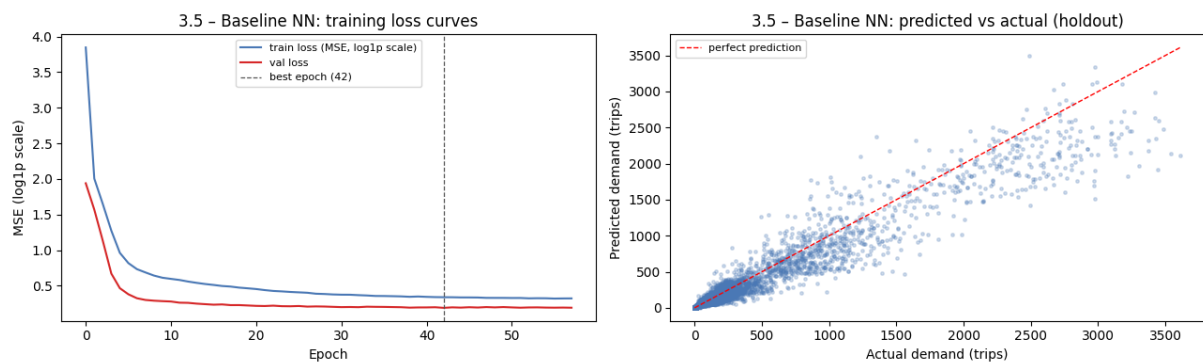


Figure 12: Model result - Neural network baseline model

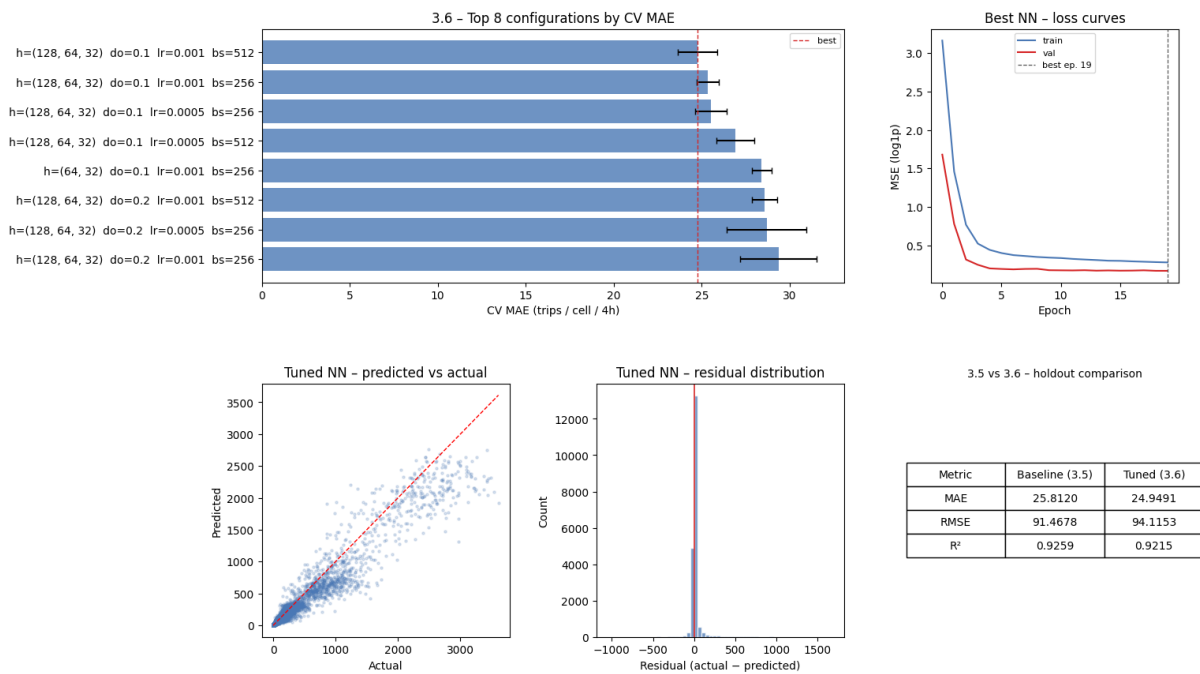


Figure 13: Model result - Neural network tuning model

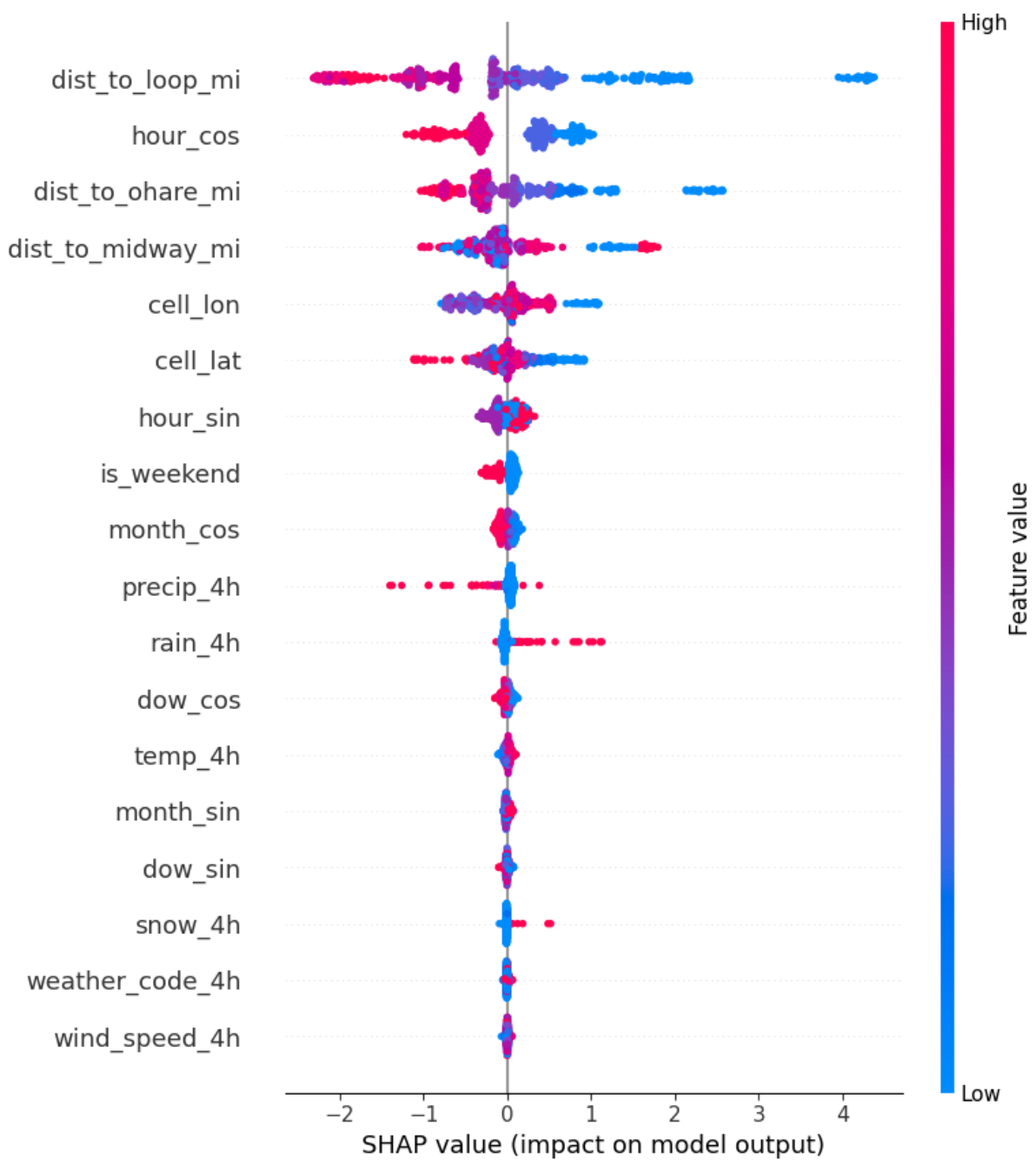


Figure 14: SHAP feature importance